# EXPERIMENT 13

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

## Handling Missing and Inappropriate Data

**Aim:** Remove duplicates, correct invalid values and fill missing values.

## Python Program

```
import pandas as pd
import numpy as np
df=pd.read_csv('Hotel_Dataset.csv').drop_duplicates()
df.loc[~df['Rating(1-5)'].between(1,5),'Rating(1-5)']=np.nan
df.loc[df['NoOfPax']<1,'NoOfPax']=np.nan
df.loc[df['Bill']<0,'Bill']=np.nan
for c in df.select_dtypes('number'):
    df[c]=df[c].fillna(df[c].median())
for c in df.select_dtypes('object'):
    df[c]=df[c].fillna(df[c].mode()[0])
print(df)
```

## Result

Duplicate rows are removed, invalid values are treated as missing, and missing values are filled.

**Note:** For CSV programs, keep the named CSV file in the same folder and check that column names match.